PHYS–467 Machine Learning for Physicists
Assignment 3: Predicting Atomization Energies with Neural Nets

Grading

Question 4

-1.0 wrong initialization
-1.5 too many layers
-3.0 completely wrong width (e.g., 1 instead of 1000)

Question 5

-0.5 not commenting on the behaviours of train and validation losses
-1.0 missing plot with losses
-1.0 choice of learning rate using fewer epochs
-4.0 not reinitialising the model while varying learning rate

Question 6

-0.5 wrong plot
-0.5 missing rmse (with correct unnormalised mse)
-1.0 wrong rmse (e.g., wrong loss normalisation)
-1.0 missing unnormalised error
(not penalised) units of rmse and mse!

Question 7

-0.5 wrong substitution (coding error), not perfect fit (not optimal choice of parameters)
-0.75 the test error with 4000 training points is missing (but the fit was ok)
-1.0 wrong function (e.g., exponential) but correct fit in log scale
-2.0 wrong function, wrong fit, unrealistic test error
-4.0 not changing model while varying n
-5.0 dimensional mismatch, e.g, missing .squeeze() (with increasing unrealistic validation/test errors)

Bonus

between 0.0 (nothing) and 1.0 (very good)